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FORLENS: A Lightweight AI-Driven Endpoint Security Solution for Real-Time Threat Detection in Small and Medium Enterprises

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Abstract— Small and Medium Enterprises (SMEs) constitute of more than 90% of businesses worldwide and are increasingly becoming a target for ransomware, insider abuse, credential theft, and lateral movement campaigns. Even though such cyber risks exist, most SMEs are not able to procure, invest or implement Security Operations Center (SOC) infrastructures due to high licensing costs, operational complexity, and hardware limitations. The endpoint security solutions that are existing in the market focuses on either detection, forensic integrity, or automated response in isolation, which has created a fragmented defensive ecosystem in organizations. This fragmented approach has proved to fail in the real-world scenarios considering the dynamic nature of today's cyber criminals. Furthermore, traditional blockchain-based forensic systems are inherent with high storage overhead and verification latency, while black-box Artificial Intelligence (AI) lack the trust among non-technical administrators.

This paper presents FORLENS, a lightweight AI-driven endpoint security ecosystem specifically designed for resource-constrained SME environments. FORLENS integrates four major defensive pillars. First, a resilient endpoint monitoring agent with embedded SMB deception, automated signature-driven containment, and watchdog-based self-healing. Second, a hybrid Machine Learning (ML)-driven Threat Intelligence Engine that correlates local behavioral anomalies with Open-Source Intelligence (OSINT) feeds. Third, LogStamping, a blockchain-anchored tamper-evident logging framework utilizing grouped SHA-256 hash aggregation over a private Ethereum Proof-of-Authority (PoA) network [1]. Finally, an Explainable Artificial Intelligence (XAI)-driven Security Orchestration, Automation, and Response (SOAR) interface optimized for non-expert operators.

Experimental evaluation conducted using CICIDS2017, UNSW-NB15, and Sysmon-based behavioral telemetry demonstrates that FORLENS maintains a low operational footprint of 2–4% CPU usage and 60–90 MB memory consumption while achieving 89.4% precision, 91.2% recall, and 90.3% F1-score in anomaly detection. The LogStamping mechanism reduces blockchain storage overhead by approximately 95% compared to naïve per-entry blockchain anchoring strategies. Additionally, integrated OSINT enrichment reduces false positive alerts by 33%, while the XAI-assisted response interface reduces Mean Time To Respond (MTTR) for non-technical administrators from 30 minutes to under 5 minutes.

The findings demonstrate that enterprise-grade defensive principles—including cryptographic forensic integrity, deception engineering, explainable autonomous response, and AI-driven behavioral analytics—can be adapted into an

affordable and scalable cybersecurity ecosystem suitable for SMEs.

Keywords: *SME Cybersecurity, Endpoint Security, Blockchain Forensics, Explainable AI, Threat Intelligence, Cyber Deception, Security Automation, Ethereum PoA, LSTM Autoencoder.*

I. INTRODUCTION

A. SME Cybersecurity Landscape

Companies worldwide are experiencing swift digitalization changes to improve and render their business processes considerably more efficient. These digital advancements have turned Small and Medium Enterprises (SMEs) into tightly linked entities that rely heavily on cloud platforms, remote teamwork, endpoint devices, and digital finance solutions. Although digital transformation has boosted operational efficiency, it also introduces its own hazards and makes organizations vulnerable to cyber-attacks. Recent cybersecurity studies indicate that SMEs are subject to a greater incidence of ransomware, phishing attempts, insider threats, and lateral-movement intrusions.

Large enterprises operate Security Operations Centers (SOCs) to counter cyber-attacks. SMEs lack the ability to run SOCs because of financial, operational and personnel constraints. Unlike large firms, SMEs do not possess specialized staff, substantial budgets, or sophisticated IT infrastructure. **Endpoint Detection and Response (EDR) tools like CrowdStrike Falcon, Microsoft Sentinel, Splunk Enterprise Security, and Palo Alto Cortex XDR** deliver enhanced and deeper threat visibility [31], [32], [34], [36]. These solutions demand higher licensing fees, cloud connectivity, and highly qualified operators, which SMEs do not have.

B. Problem Statement and Constraints

SMEs employ fragmented security strategies that lack integration among systems and tools, forensic integrity, automated response, and behavioral analysis. This fragmentation has produced gaps that adversaries are increasingly exploiting.

Threat actors are recognized for turning off security tools, wiping log files and tampering with forensic data to impede investigations. An increase in advanced attack methods like ransomware-as-a-service (RaaS), Living-off-the-Land (LotL) attacks, credential dumping, and stealthy lateral movement.

Traditional centralized logging mechanisms are susceptible to alteration since attackers may obtain admin



1



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rights that allow them to change or erase evidence [2]. Blockchain-based logging solutions aim to address this problem by ensuring immutability, but they entail greater storage expenses and increased computational load.

Simultaneously, incorporating Artificial Intelligence (AI) into cybersecurity workflows presents its own difficulty, even though Machine Learning (ML) tools enhance detection capabilities, they operate as black boxes. Decision-making and alert-analysis logic are hidden from users. Users cannot discern the reasoning patterns of these systems. Non-technical administrators who are often common on SMEs find it hard to understand technical feedback and are unable to trust AI systems completely for automated decision-making scenarios.

These constraints drive the requirement for an integrated, lightweight, understandable, and tamper-proof endpoint security solution designed expressly for SMEs.

33 C. Contributions of FORLENS

This study presents FORLENS, a lightweight AI-powered endpoint security platform crafted to overcome the operational, financial, and technical constraints experienced by Small and Medium Enterprises (SMEs). In contrast to traditional cybersecurity tools that concentrate on separate defensive functions like malware detection, centralized logging, or automated response on their own, FORLENS offers a cohesive and integrated architecture that merges blockchain-based forensic integrity, behavioral analytics, cyber deception, Explainable Artificial Intelligence (XAI), and autonomous orchestration into one operational framework.

3 One of the main contributions of this study is the creation of the LogStamping forensic auditing framework, which presents a grouped blockchain anchoring approach for scalable and tamper-proof forensic preservation. Current blockchain-based logging solutions often experience rapid storage expansion and synchronization costs because each log entry is individually written to the blockchain ledger. FORLENS overcomes this drawback by aggregating SHA-256 digests in groups, merging multiple telemetry records into cryptographically verifiable log clusters prior to blockchain synchronization. This approach markedly lowers on-chain storage demands while maintaining forensic integrity and a deterministic tamper-detection capability. Incorporating a private Ethereum Proof-of-Authority (PoA) consensus further enhances scalability and operational efficiency for SME settings by cutting computational load and transaction latency relative to traditional blockchain consensus mechanisms.

Another significant contribution of FORLENS is the integration of lightweight cyber deception directly into endpoint systems. Traditional deception frameworks often depend on dedicated honeypot infrastructures that require substantial hardware resources, operational maintenance, and specialized deployment expertise. In contrast, FORLENS introduces host-level SMB deception through lightweight lure shares embedded directly within monitored endpoints. This approach enables proactive lateral movement detection without requiring separate deception infrastructure. By monitoring unauthorized interactions with decoy SMB resources, the framework provides early-warning visibility into adversarial reconnaissance and lateral traversal activity while maintaining low operational overhead suitable for resource-constrained environments.

The study additionally offers a combined AI-driven behavioral analytics platform that can identify both familiar and novel attack patterns by merging statistical anomaly detection with sequential behavior modeling. The architecture merges Isolation Forest techniques with Long Short-Term Memory (LSTM) Autoencoder models to examine endpoint telemetry and spot divergences from defined behavioral baselines. This blended approach facilitates the identification of advanced malicious actions such as ransomware processes, PowerShell misuse, credential exfiltration, persistence creation, and fileless malware operations. In contrast to traditional signature-based solutions that depend solely on preset Indicators of Compromise (IOCs), the FORLENS behavioral engine constantly adjusts to shifting operational behavior trends, thereby enhancing insight into new and previously unknown threats.

Another contribution of this research is the incorporation of Open-Source Intelligence (OSINT) enrichment into the behavioral detection workflow. Current anomaly detection platforms often produce a high number of false alarms due to a lack of contextual knowledge about external threat intelligence. FORLENS tackles this issue by matching behavioral irregularities with worldwide threat intelligence sources such as MISP, AlienVault OTX, AbuseIPDB, VirusTotal, and OpenCTI. This contextual threat linking enhances alert ranking, improves the precision of risk scoring, and markedly cuts down false positive creation in SME security settings.

The FORLENS platform additionally provides an XAI-based orchestration system designed expressly for administrators without technical expertise. A key drawback of contemporary AI-enhanced cybersecurity solutions is the trust gap in operation that stems from opaque black-box machine-learning models. Alerts produced by such hidden anomaly-detection tools are frequently hard for non-specialist staff to understand, leading to slower incident response and diminished confidence in automated security processes. FORLENS tackles this issue by converting anomaly-detection results into clear, human-readable behavioral stories that outline suspicious activity trends, inferred attack tactics, and suggested remediation steps. This focus on explainability boosts administrators' comprehension, eases mental strain, and strengthens operational decision-making throughout security incidents.

A further significant outcome of this study is the deployment of robust endpoint survivability mechanisms via a watchdog-driven self-healing framework. Adversaries often try to shut down monitoring agents and forensic services during compromise activities to avoid detection and erase evidential traces. FORLENS counters this risk by continuously monitoring process integrity and providing automated service restoration functions. The watchdog component constantly checks the availability of endpoint agents and automatically relaunches any services that have been tampered with or terminated. This self-repair approach guarantees uninterrupted telemetry collection and maintains forensic continuity even amid active hostile interference.

The framework additionally provides a lightweight autonomous orchestration and response architecture able to support real-time containment actions in low-resource settings. The orchestration engine combines behavioral analytics, deception-triggered events, threat-intelligence

correlation, and contextual confidence scoring to automate repetitive security tasks such as process termination, file quarantine, IOC blocking, endpoint isolation, forensic snapshot creation, and blockchain synchronization. In contrast to enterprise-scale SOAR solutions that often demand dedicated infrastructure and extensive configuration expertise, the FORLENS orchestration framework was deliberately engineered for operational simplicity and SME-friendly deployment.

From a scalability perspective, FORLENS offers a modular cybersecurity architecture that can function across decentralized and partially linked environments. The incorporation of offline telemetry buffering, asynchronous synchronization mechanisms, distributed forensic verification, and lightweight AI inference allows the framework to preserve operational continuity even during intermittent connectivity conditions commonly found in SME infrastructures.

This study also helps close the cybersecurity capability divide between large enterprises and SMEs. Current enterprise-grade EDR, SIEM, and SOAR solutions are frequently out of reach for SMEs because of cost, technical, and operational obstacles. FORLENS shows that sophisticated cybersecurity ideas—including blockchain-assisted forensic auditing, behavioral AI analytics, cyber deception, Explainable AI, and autonomous orchestration can be molded into a lean, cost-effective ecosystem appropriate for resource-constrained settings while preserving defensive strength.

Finally, this study adds to the wider domain of resilient cybersecurity engineering by showing the practicality of merging forensic integrity, adaptive behavioral intelligence, deception-assisted detection, explainable orchestration, and autonomous containment into a single endpoint security framework. The experimental assessment confirms that FORLENS effectively balances detection accuracy, operational explainability, computational efficiency, and forensic scalability while staying suitable for real-world SME deployment scenarios.

Taken together, these inputs position FORLENS as an all-encompassing, scalable, interpretable, and tamper-proof cybersecurity framework that can enhance an organization's resilience to contemporary endpoint attacks while taking into account the operational constraints of resource-limited SME environments.

II. LITERATURE REVIEW

As SMEs increasingly rely on digital infrastructure, their exposure to cyber-security attacks and threats grows because the attack surface expands. Compared with larger firms, SMEs are more prone to being targeted due to limited budgets and a shortage of skilled IT personnel. Although some SMEs can allocate funds to cybersecurity, this often results in a disjointed strategy because consolidating and integrating systems is costly and impractical for them. The varieties of attacks reported by advisories continuously evolve and now encompass ransomware, insider threats, file-less malware, Advanced Persistent Threats (APTs), credential theft, and lateral movement.

A. Traditional and Advanced Endpoint Threats

Traditional endpoint protection systems are not capable to defend businesses against advanced attacks such as

polymorphic malware, zero-day exploits and behavioral attacks. This limitation is mainly due to the reliance on signature-based detection approaches. Traditional systems are unable to combat against modern adversarial tactics such as Living-off-the-Land (LotL) attacks, privilege escalation, and stealthy persistence mechanisms. As a result modern cybersecurity research has now shifted towards behavioral analytics, artificial intelligence, autonomous orchestration, deception engineering, and tamper-resistant forensic systems.

The next major obstacle is preserving the integrity of logging systems. When logs are breached and an adversary can modify them, the entire purpose of logging is defeated. To tackle this issue, blockchain technology has been employed. Blockchain has demonstrated resistance to tampering and can guarantee forensic integrity. ProvChain introduced auditing mechanisms for cloud environments [5]. It utilizes immutable transaction records to maintain forensic evidence. Nonetheless, this framework experienced considerable storage expansion and linear verification complexity, restricting its scalability in large-scale deployments. In the same vein, EngraveChain suggested blockchain-assisted digital evidence preservation with distributed verification mechanisms but created substantial computational and synchronization overhead that complicated deployment in resource-constrained settings [6].

B. Blockchain in Forensic Integrity

Blockchain-based Cloud Auditing and Logging Systems (BCALS) kept depending on frequent blockchain synchronization and increased verification latency while boosting scalability via partial off-chain storage methods [7]. Over time, the majority of current blockchain logging solutions that store each log entry on-chain have led to excessive blockchain growth, high transaction costs, and poor verification performance. For SMEs that rely on commodity hardware with limited processing capabilities, these constraints become especially difficult. Moreover, many existing blockchain forensic platforms fail to address practical deployment challenges such as low-latency verification, endpoint resilience, offline continuity, and seamless integration with endpoint security operations.

Endpoint Detection and Response (EDR) platforms have progressed rapidly alongside blockchain forensics thanks to the integration of artificial intelligence and behavioral analytics. Sophisticated telemetry inspection, automated orchestration, and unified visibility across corporate environments are delivered by commercial solutions such as CrowdStrike Falcon, Microsoft Sentinel, Palo Alto Cortex XDR, Splunk Enterprise Security, and Elastic SIEM. These tools leverage machine-learning techniques to identify malicious persistence tactics, anomalous network activity, suspicious applications, and credential misuse.

Enterprise-grade EDR solutions remain cost-prohibitive and operationally unfeasible for most small- and medium-sized enterprises, even though they are effective [37], [38]. Cloud-native architectures, dedicated Security Operations Centers (SOCs), extensive telemetry pipelines, and experienced cybersecurity personnel are typically required for their deployment. Moreover, the substantial CPU, memory, and storage demands of many commercial products exceed the capabilities of low-budget SME environments. While open-source tools such as Wazuh and OSSEC cut licensing expenses, they often lack blockchain-based forensic logging, explainable AI

integration, strong autonomous response functions, and lightweight deception engineering.

2 The advent of artificial intelligence (AI) and machine learning (ML) has fully transformed research on threat intelligence and intrusion detection. 7 Deep learning architectures such as Long Short-Term Memory (LSTM) networks, Recurrent Neural Networks (RNNs), Convolutional Neural Networks (CNNs), autoencoders, and 25 transformer-based models have demonstrated strong capability in detecting anomalous behavioral patterns within network traffic and endpoint telemetry [15], [16], [17]. Since they are able to identify statistical outliers in high-dimensional data, clustering-based anomaly detection techniques such as isolation forests have gained popularity [40], [42], [43].

Many studies have demonstrated that hybrid AI models can boost the precision of attack detection. Hybrid FFN-LSTM designs have achieved high classification rates on standard intrusion datasets such as NSL-KDD, CICIDS2017, and UNSW-NB15. 20 Nevertheless, numerous approaches incur considerable computational overhead, rendering them unsuitable for real-time use on systems with limited endpoints. Deep neural networks often demand extensive memory, large training sets, and GPU acceleration, which makes them ill-suited for low-power devices.

Even though they require lower processing overhead, lightweight AI approaches such as statistical anomaly detection frameworks and shallow Feedforward Neural Networks (FFNs) can occasionally compromise detection depth and contextual insight. A core unresolved problem in endpoint security studies remains the balance between computational efficiency and detection precision.

Another major difficulty linked to AI-driven cybersecurity solutions concerns explainability and operational trust. 27 Numerous machine learning models function as opaque black-box tools that deliver classifications without clear justification. Non-technical administrators frequently find it hard to interpret anomaly scores, behavioral categorizations, or confidence metrics produced by AI systems. This opacity undermines confidence in autonomous response mechanisms and prolongs incident response times.

C. AI and Explainability in Cybersecurity

37 Consequently, explainable artificial intelligence (XAI) has emerged as an important research field within cybersecurity operations. In order to improve operators' understanding of AI-driven choices, techniques such as SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-Agnostic Explanations), attention-based visualisation, and 6 rule-based narrative generation have been suggested [21], [22], [23], [25]. Studies indicate that explainability substantially diminishes cognitive load, boosts analyst confidence, and shortens Mean Time To Respond (MTTR) in incident investigations [26], [29]. Nonetheless, most existing XAI tools continue to target highly skilled analysts instead of non-technical SME administrators.

D. Cyber Deception and Threat Intelligence

As an anticipatory defense measure capable of spotting enemy scouting and lateral movement before production systems are jeopardized, cyber deception has garnered considerable interest. To mislead attackers and increase their uncertainty, conventional deception architectures chiefly depend on honeypots, honeytokens, fake credentials, decoy

services, and fabricated vulnerable configurations [47], [51], [54].

Although they provide extensive forensic insight, high-interaction honeypots require dedicated infrastructure, considerable processing capacity, and continuous maintenance. Although low-interaction deception solutions reduce operational load, they typically suffer from weak integration with automated response systems and limited visibility. SMB deception serves as an effective early-warning mechanism in corporate environments, as many studies have demonstrated that adversaries often engage with SMB shares during lateral-movement activities [52].

However, most deception systems currently deployed operate independently of endpoint detection frameworks and are not incorporated with autonomous orchestration, machine-learning-based analytics, or forensic auditing. Moreover, the literature still lacks studies on lightweight deception solutions designed for SMEs.

Threat Intelligence Platforms (TIPs) combined with Open-Source Intelligence (OSINT) aggregation utilities are increasingly important for contextual threat analysis. Tools such as MISP, AlienVault OTX, AbuseIPDB, VirusTotal, and OpenCTI enable organizations to compare internal telemetry with global indicators of compromise, hostile infrastructure patterns, and adversary tactics [68], [69], [70]. Integrating OSINT feeds into on-premise behavioral analytics improves contextual insight and reduces false-positive rates by enriching alerts with external intelligence.

Even with these improvements, current Threat Intelligence frameworks often produce too many alerts, uneven scoring methods, and operational intricacy that swamp small IT teams. Many solutions focus on large-scale enterprises and do not offer lean contextual enrichment tools tailored for SME settings.

Research papers also indicate a growing emphasis on autonomous orchestration and Security Orchestration, Automation, and Response (SOAR) frameworks [63], [64], [66]. Modern SOAR solutions automate everyday security tasks, such as IOC blocking, shutting down malicious processes, collecting evidence, and generating tickets. Combining reinforcement learning with self-adjusting policies has further boosted the adaptability of automated security operations.

However, numerous SOAR platforms still rely on centralized cloud infrastructure and demand considerable configuration expertise. In addition, completely autonomous security solutions provoke worries about explainability, accidental containment measures, and overreliance on AI-driven decisions.

Many important research gaps therefore remain unresolved in the current literature. First, existing blockchain forensic methods continue to face scalability limitations and inefficient storage techniques. Second, commercial EDR and SOAR products are still prohibitively expensive and operationally complex for small- and medium-sized enterprises. Third, lightweight deception frameworks rarely integrate autonomous response, forensic integrity, and explainable AI components. Fourth, AI-driven security platforms frequently lack operational transparency for administrators lacking extensive expertise. Finally, most research focuses on single defensive

components rather than on integrated architectures that can deliver comprehensive resilience against modern endpoint threats.

These constraints collectively demonstrate the need for a lightweight, transparent, tamper-proof, and seamlessly integrated endpoint security solution designed for SME environments. The FORLENS framework addresses these gaps by combining blockchain-backed forensic integrity, host-level deception engineering, Explainable AI-driven orchestration, robust endpoint monitoring, and hybrid behavioral analytics into a single cybersecurity architecture.

III. METHODOLOGY

A. Modular Architecture Design

The methodology employed in this study centers on the design, implementation, integration, and assessment of a lightweight AI-powered endpoint security ecosystem deliberately tailored for Small and Medium Enterprise (SME) settings. The FORLENS framework was built with a modular security architecture that merges behavioral analytics, blockchain-supported forensic auditing, deception engineering, Explainable Artificial Intelligence (XAI), and automated response orchestration into a cohesive cybersecurity ecosystem.

The study's approach comprised five key phases: designing the system architecture, creating the endpoint agent, implementing a blockchain forensic framework, integrating AI-driven threat intelligence, and conducting experimental evaluation. Every phase was crafted to overcome particular operational drawbacks of current SME cybersecurity tools, all the while keeping computational load minimal and ensuring the solution can be deployed in environments with limited resources.

The first stage of the study consisted of examining the functional constraints of conventional endpoint security solutions and pinpointing the key protection weaknesses impacting SMEs. Current enterprise-level Endpoint Detection and Response (EDR) platforms, blockchain forensic tools, and self-directed orchestration frameworks were assessed to gauge their advantages, drawbacks, scalability limits, and implementation needs. From this assessment, a modular architecture was created to guarantee low-overhead performance while preserving compatibility among forensic auditing, behavioral analytics, deception-driven detection, and automated response components.

The FORLENS framework was built as a tiered ecosystem that includes endpoint telemetry gathering, lightweight deception units, behavioral analytics processors, forensic auditing backed by blockchain, threat-intelligence enrichment, and self-directed orchestration layers. This modular design allows each component to function on its own while feeding telemetry and contextual insights into a unified orchestration pipeline.

B. Resilient Endpoint Agent

The endpoint monitoring system was built as a slim, multithreaded Python agent designed for minimal resource usage in SME settings. Python was chosen because its broad ecosystem provides machine-learning packages, system-level monitoring APIs, cryptographic tools, and fast development potential. The agent constantly watches essential system

actions such as process start and stop, file-system activity, network traffic, authentication occurrences, and Windows Security logs. These monitoring functions are realized by linking to native OS telemetry sources and Sysmon-derived event logs, providing full insight into endpoint behavior.

To enable real-time behavioral observation, the agent uses an asynchronous, event-driven framework together with a lightweight, queue-based telemetry pipeline. Every monitoring component - process, file, network, and honeypot detection is run in its own thread, permitting parallel data gathering without hindering execution. This multithreaded approach guarantees ongoing monitoring across various system layers while preserving responsiveness and stability.

The agent also employs selective event-filtering techniques to cut down noise and prevent needless handling of harmless system actions. It captures only pertinent events and sends them for analysis, reducing superfluous system calls and decreasing computational load. Moreover, hash-based detection is activated solely during process-creation events, wherein executable files are examined via SHA-256 hashes and matched against a predefined Indicator of Compromise (IOC) database. This focused method removes the requirement for continual file scans, markedly boosting performance efficiency.

To enhance detection capabilities, a lightweight deception mechanism is embedded within the agent as an SMB honeypot. The system watches Windows Security Event ID 5140 to spot unauthorized attempts to access fake network shares, allowing early detection of lateral movement and reconnaissance. Furthermore, the agent incorporates an automated response feature that carries out immediate containment steps, like isolating malicious files when they are found.

Resilience and continuity are provided by a watchdog-driven self-healing system that constantly watches the agent process and automatically relaunched it when it stops. Moreover, the agent can operate offline by locally queuing telemetry data during network outages and later synchronizing it with the backend once connectivity is restored. Together, these design choices ensure that the endpoint agent delivers high availability, minimal resource consumption, and reliable threat detection, making it suitable for deployment in resource-constrained SME environments.

So overall, the lightweight design of the endpoint agent was achieved through efficient resource management strategies, including short monitoring intervals, selective event filtering, and avoidance of redundant system calls. A modular thread-based architecture ensures that process, file, network, and honeypot monitoring modules operate independently without blocking execution. Furthermore, the agent minimizes computational overhead by performing hash-based detection only during process creation events, rather than continuously scanning the system. This design allows the agent to maintain real-time monitoring capabilities while keeping CPU and memory usage within acceptable limits for SME environments.

C. LogStamping and Blockchain Forensics

The forensic auditing methodology employed in FORLENS was largely driven by the requirement to harmonize cryptographic soundness, scalability, and operational effectiveness in environments with limited

resources. Conventional blockchain-based logging designs often encounter rapid storage expansion because raw log data are written directly onto the ledger. To overcome this issue, the suggested methodology introduced a grouped hash-anchoring approach where endpoint telemetry records are combined into configurable log batches before being submitted to the blockchain. Each group was fingerprinted using SHA-256 cryptographic hashing, and only the resulting digest was committed to the blockchain. This reduced the blockchain storage complexity from linear per-entry growth to grouped digest growth proportional to $O(N/G)$, where N represents total log entries and G represents configurable group size.

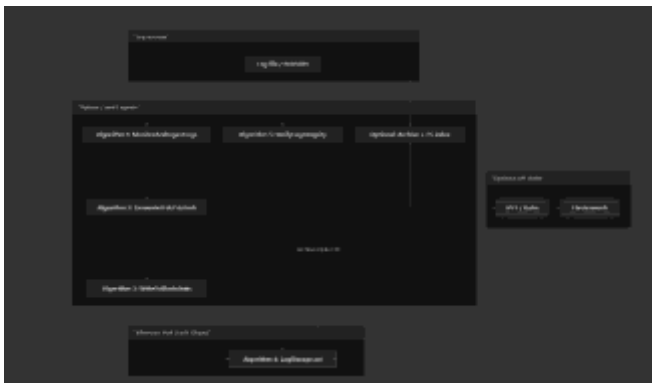


Fig. 1. End-to-end system architecture: log sources feed the Python-based ingestion agent, which computes SHA-256 group digests and submits them to the Ethereum PoA smart contract. Verified logs are optionally archived to IPFS and indexed in Elasticsearch.

The ingestion methodology was crafted with a dual-condition flush design to handle both low-throughput and high-throughput telemetry settings. Endpoint telemetry events were constantly queued into a temporary logGroup container. The group was closed when either the preset line limit G was met or a configurable ingest timeout passed. This two-trigger system guaranteed that forensic anchoring stayed agile during periods of sparse telemetry while also spreading transaction costs during bursts of rapid logging. This strategy avoided endless buffering pauses and kept blockchain transaction rates within set bounds across fluctuating workloads.

To guarantee deterministic forensic replay and integrity verification, every telemetry entry within each group was merged in its original sequential order before digest generation. The resulting byte stream was encoded in UTF-8 and run through the SHA-256 hashing algorithm to produce a fixed-size 256-bit digest expressed as a hexadecimal string. This approach ensured that any byte-level alteration including character substitution, whitespace changes, line insertion, deletion, or entry reordering generated a cryptographically distinct digest with overwhelming likelihood. Consequently, the framework achieved deterministic tamper detection while preserving the original event ordering.

The blockchain communication methodology utilized a private Ethereum network operating under the Clique Proof-of-Authority (PoA) consensus mechanism [8]. PoA consensus was selected because it provides deterministic block creation, low energy consumption, reduced computational overhead, and near real-time transaction finality without requiring Proof-of-Work mining operations [8]. The ingestion agent interacted with the blockchain through the web3.py RPC interface, constructing and signing transactions invoking the Solidity-

based storeLogHash() smart contract function. Grouped digests, timestamps, sequential indices, and optional IPFS references were committed on-chain to establish immutable forensic anchors suitable for later verification and audit replay.

The smart contract approach employed append-only semantics to maintain an immutable forensic sequence. The deployed Solidity contract kept indexed mappings linking sequential audit identifiers with grouped SHA-256 digests. Once recorded, blockchain entries could not be altered or removed because of the Ethereum Virtual Machine's execution guarantees. Additional existence-check functions were added to enable efficient integrity verification and digest lookup during forensic reconstruction.

To improve long-term forensic scalability, the approach utilized encrypted off-chain storage through the InterPlanetary File System (IPFS). Verified telemetry collections were protected with AES-256 symmetric encryption prior to storage. The resulting Content Identifier (CID) from IPFS was subsequently associated with the corresponding blockchain digest record. This integrated architecture preserved cryptographic integrity while markedly reducing blockchain size expansion and synchronization load. Separating on-chain integrity proofs from off-chain encrypted payloads enabled scalable long-term forensic archiving without embedding raw telemetry in the chain.

The integrity verification approach recreated grouped telemetry during forensic replay, using the identical grouping policy applied at ingestion. Verification clusters were rebuilt either based on line-count limits or by reconstructing timestamp intervals with the verification timeout.

Recomputed SHA-256 digests were compared against blockchain-anchored records retrieved through indexed smart-contract queries. Any mismatch immediately marked the relevant telemetry batch as modified. Deliberate tampering trials that inserted, deleted, swapped, or reordered lines achieved a 100% detection rate, confirming the grouped-hash method.

The framework additionally incorporated Elasticsearch indexing to enable rapid forensic extraction and behavioral analysis of verified telemetry archives. Once the integrity validations succeeded, the verified telemetry collections were loaded into Elasticsearch, allowing full-text queries, date-range filtering, anomaly connections, and quick incident rebuilding. This strategy allowed FORLENS to combine blockchain-based forensic integrity with an enterprise-grade forensic search capability while maintaining operational scalability suitable for SME deployments.

Experimental benchmarking approach was crafted to assess blockchain storage efficiency, verification delay, tamper-detection precision, and ingestion scalability across different telemetry volumes. Comparative tests were performed against BCALS, ProvChain, and EngraveChain designs using data sets spanning 2,000 to 12,000 log entries [2], [5], [6]. Findings revealed that grouped hash anchoring markedly lowered blockchain storage expansion while keeping verification delay within acceptable limits. With 10,000 log entries, the framework consumed roughly 0.62 MB of blockchain storage versus 12.5 MB for EngraveChain and 10 MB for ProvChain, confirming the scalability benefits of grouped digest anchoring.

TABLE I

QUANTITATIVE COMPARISON OF STORAGE AND VERIFICATION TIME AT 10,000 LOG ENTRIES

System	Verification Time (ms)	Storage (MB)	Real-Time Support
LogStamping v1.0	≈ 155,000	0.62	High
BCALS [2]	≈ 114,818	1.0	Limited
EngraveChain [3]	≈ 221,000	12.5	Limited
ProvChain [4]	≈ 209,000	10.0	Not Emphasized

D. AI-Driven Behavioral Detection

The AI-driven behavioral detection approach employed in FORLENS aims to create a lightweight, flexible, and interpretable anomaly-detection system that can spot advanced endpoint threats instantly while keeping computational demands low enough for SME environments. Conventional endpoint security solutions largely rely on fixed signature libraries and rule-based detection mechanisms, which frequently fail to stop zero-day exploits, polymorphic malware, fileless attacks, Living-off-the-Land (LotL) tactics, and stealthy ransomware campaigns. To address these shortfalls, the suggested approach concentrates on characterizing typical endpoint activity and spotting anomalies that signal malicious behavior.

The behavioral analytics framework employed a mixed detection approach that merges statistical anomaly spotting, sequential behavior modeling, contextual threat-intelligence linking, and generation of explainable inferences. The goal of the design was to enhance detection thoroughness while reducing false alerts and maintaining operational clarity for administrators without technical expertise.

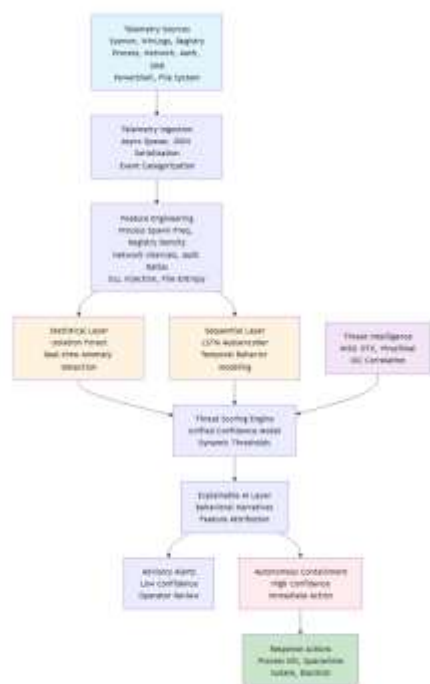


Fig. 2. Step by Step Conversion Process

Collected events were transformed into structured telemetry objects via JSON serialization prior to feeding into the behavioral analytics pipeline.

To boost real-time performance and cut processing delay, the telemetry ingestion framework employed an asynchronous queue-based event stream. Events were sorted by behavioral severity and operational relevance before analysis. High-priority telemetry comprised encoded PowerShell execution, fast file-change sequences, suspicious parent-child process chains, unauthorized registry persistence modifications, credential-dumping signs, SMB enumeration activity, irregular network beaconing, privilege-escalation attempts, remote thread-injection actions, and process-hollowing signs.

The approach employed comprehensive feature-engineering steps to convert raw endpoint telemetry into behavioral representations suitable for machine learning. Feature extraction aimed at producing statistical as well as temporal depictions of endpoint activity patterns. The resulting feature vectors comprised process spawn frequency, execution duration, command-line entropy, registry modification density, network connection intervals, SMB traversal frequency, authentication success and failure ratios, DLL injection indicators, file entropy variance, parent-child process lineage, privilege transition sequences, and API call frequency distributions.

To maintain analytical uniformity, the preprocessing steps comprised scaling, encoding of categorical variables, aligning timestamps, interpolating missing data, filtering noise, and segmenting time series with a sliding window.

The first stage of the behavioral detection methodology utilized Isolation Forest algorithms for lightweight statistical anomaly detection. Isolation Forest was selected because of its computational efficiency, scalability, and suitability for high-dimensional telemetry environments. The algorithm operates by recursively partitioning feature space into randomized decision trees where anomalous behaviors become isolated using fewer partition operations than normal observations.

This approach allowed the framework to detect statistically unusual endpoint behaviors without needing large labeled attack datasets. Anomaly confidence scores produced by the Isolation Forest algorithm were constantly refreshed using dynamic behavioral baselines taken from endpoint activity histories.

To enhance robustness against shifting operational patterns, adaptive baseline learning mechanisms were deployed. Static behavioral profiles frequently generate many false positives because of routine organizational workflow changes, software updates, and variations in user activity. Consequently, baseline profiles were regularly readjusted through incremental retraining intervals while maintaining historical anomaly weighting to reduce the risk of adversarial poisoning.

While statistical anomaly detection efficiently spots isolated deviations, numerous advanced attacks appear chiefly as a sequential behavioral progression rather than single-event irregularities. Consequently, the approach employed Long Short-Term Memory (LSTM) Autoencoder networks to capture temporal behavioral dependencies across endpoint activity sequences [9], [10], [13], [14].

The LSTM Autoencoder model included encoder and decoder parts that were taught to rebuild regular operational sequences [10]. The encoder condensed behavioral telemetry windows into latent feature vectors that capture the temporal characteristics of endpoint activity, and the decoder regenerated the anticipated behavior from these condensed representations.

Reconstruction loss functioned as the primary anomaly indicator for sequential behavior analysis. A higher reconstruction error signaled irregular activity sequences that deviated from the learned operational patterns. This approach allowed the system to detect multi-stage attack tactics such as ransomware encryption processes, PowerShell automation pipelines, persistence-building sequences, credential-stealing operations, lateral SMB movement, coordinated process creation, and automated reconnaissance activities.

Temporal sequence construction employed sliding-window segmentation to maintain contextual continuity across consecutive events. Behavioral windows encompassed ordered process relationships, authentication sequences, registry modification timelines, network communication bursts, SMB interaction chains, and file access progression patterns.

The sequential learning approach was deliberately tuned to keep inference latency low on SME hardware platforms. Compact network designs and limited hidden-layer sizes were chosen to strike a balance between modeling depth and computational efficiency.

To enhance contextual insight and operational usefulness, the behavioral analytics engine incorporated Open-Source Intelligence (OSINT) enrichment capabilities. Anomalous behaviors detected by the statistical and sequential detection tiers were matched with external threat-intelligence sources such as MISP, AlienVault OTX, AbuseIPDB, VirusTotal, and OpenCTI.

Threat intelligence correlation integrated malicious IP addresses, command-and-control infrastructure cues, recognized ransomware hashes, dubious domains, adversary tool fingerprints, IOC confidence metrics, and threat-actor associations. This contextual enrichment approach markedly enhanced threat ranking and cut down false positives produced by stand-alone anomaly detection systems.

The threat scoring system merged several analytical facets into a single behavioral confidence framework. The resulting threat ratings integrated Isolation Forest anomaly confidence, LSTM reconstruction loss, IOC correlation strength, behavioral severity weighting, historic endpoint reputation, SMB deception trigger events, process trust classification, and threat-intelligence reputation scores.

Weighted scoring limits were automatically modified in line with operational sensitivity policies to balance detection aggressiveness with alert fatigue.

To further strengthen detection reliability, the approach integrated deception-assisted behavioral correlation. Interactions with SMB lure shares produced higher contextual risk weighting in the scoring pipeline. Because legitimate users seldom access decoy resources, SMB lure engagement markedly raised confidence that the related actions signified malicious reconnaissance or lateral movement behavior.

E. XAI and Autonomous Orchestration

The Explainable Artificial Intelligence (XAI) approach was incorporated straight into the behavioral inference pipeline to overcome the operational trust issues linked to black-box machine learning models. Rather than presenting obscure anomaly probabilities or numeric confidence scores, the XAI component produced human-readable behavioral narratives that outlined suspicious activity patterns, deduced attack behaviors, and suggested containment measures.

The produced explanations contained contextual results like irregular PowerShell runs that use encoded commands and show privilege-escalation actions, swift rises in file entropy suggesting possible ransomware encryption, and step-by-step SMB enumeration patterns that align with lateral-movement scouting.

The explanation engine also emphasized the telemetry attributes that most heavily influence anomaly classification, thereby enhancing administrators' comprehension and confidence in their response.

The behavioral detection engine was closely coupled with the self-directed orchestration and containment system to enable instantaneous defensive action. When threat actions surpassed set confidence limits, they automatically launched containment procedures such as terminating processes, quarantining files, blacklisting IOCs, isolating networks, ending SMB sessions, containing endpoints, creating forensic snapshots, and synchronizing blockchain forensics.

Confidence-aware escalation policies were introduced to avoid over-automation when detections have low certainty. Low-confidence anomalies produced advisory alerts that needed operator confirmation, while high-confidence ransomware or active compromise signals caused instant autonomous containment.

The experimental assessment approach evaluated the behavioral detection engine by employing simulated VM environments to examine the FORLENS platform's detection performance. Additional adversarial simulations were performed using Metasploit and various other attack tools.

The assessment criteria comprised precision, recall, F1-score, ROC-AUC, detection latency, false positive rate, CPU usage, memory usage, Mean Time To Detect (MTTD), Mean Time To Respond (MTTR), and sequential anomaly reconstruction accuracy.

Experimental findings showed that the combined behavioral analytics approach effectively harmonized detection precision, situational awareness, interpretability, and processing efficiency. By merging statistical anomaly detection, sequential behavior modeling, threat-intelligence enrichment, deception-driven correlation, and Explainable AI, a robust and operationally feasible behavioral defense framework was created, tailored specifically for SME cybersecurity settings.

FORLENS's AI-powered security agent was built to act as a smart autonomous helper that can aid real-time threat analysis, contextual alert interpretation, and incident-response coordination in SME settings. It constantly observed behavioral telemetry from endpoint devices and applied machine-learning-based anomaly scoring, threat-intel correlation, and contextual risk assessment to spot dubious actions. Instead of typical alert systems that produce raw

technical data, the AI agent converted identified anomalies into easy-to-read explanations, helping non-technical administrators grasp the situation. It also supported automated decision-making by suggesting suitable containment steps like terminating processes, isolating endpoints, blocking IOCs, and preserving forensic evidence. By combining Explainable AI (XAI), behavioral analytics, and autonomous orchestration workflows, the AI agent boosted detection efficiency, cut incident-response time, and strengthened overall cyber-security resilience in resource-limited SME infrastructures. The LLM employed for this task is TinyLlama-1.1B-Chat-v1.0, which was trained on cybersecurity best practices and non-technical guidance

IV. RESULTS AND DISCUSSION

An experimental assessment of the FORLENS framework was performed to gauge its efficacy, scalability, forensic soundness, ability to detect behavior, resource efficiency, and operational appropriateness in Small and Medium Enterprise (SME) environments. The study aimed to confirm if the suggested lightweight cybersecurity ecosystem can provide enterprise-level protection while keeping computational demands low enough for standard commodity hardware

A. Detection Performance and OSINT Enrichment

The test setup comprised several Windows 10 client machines featuring Intel Core i5 CPUs and 8 GB of RAM, all linked to a private Ethereum Proof-of-Authority (PoA) blockchain that ran on Ubuntu-based validator nodes. The setup reproduced typical SME operational scenarios, such as telemetry from multiple endpoints, flaky network connectivity, ransomware incidents, insider attacks, credential misuse, and lateral-movement vectors.

The behavioral analytics trials employed standard intrusion detection datasets such as CICIDS2017, UNSW-NB15, TON IoT, along with bespoke Sysmon telemetry gathered from adversary simulations. Attack scenarios were carried out with Atomic Red Team, Metasploit, Sliver C2, PowerShell Empire, and Caldera to mimic authentic adversary actions, encompassing ransomware rollout, privilege escalation, PowerShell misuse, SMB-driven lateral movement, credential extraction, and the establishment of persistence.

The behavioral analytics engine showed robust anomaly detection ability across both statistical and sequential attack scenarios. The combined Isolation Forest and LSTM Autoencoder design delivered strong detection results while keeping inference latency low enough for real-time endpoint monitoring. Tests yielded an overall precision of 89.4 %, recall of 91.2 %, and an F1-score of 90.3 % on the tested attack datasets. The LSTM Autoencoder's sequential modeling markedly enhanced identification of ransomware encryption processes, coordinated process execution chains, and lateral movement activities that would be hard to spot with standalone statistical analysis alone.

Incorporating Open-Source Intelligence (OSINT) enrichment further enhanced situational awareness and lowered the creation of false positives. Linking with MISP, AlienVault OTX, AbuseIPDB, VirusTotal, and OpenCTI allowed the system to verify doubtful IP addresses, harmful domains, and recognized ransomware signals using outside threat-intel sources. Tests showed that OSINT enrichment

cut false-positive alerts by roughly 33 % versus standalone anomaly detection lacking contextual linkage.

B. Deception Effectiveness and Self-Healing

The deception-assisted detection approach proved highly effective in lateral-movement exercises. Lightweight SMB lure shares installed on the observed endpoints consistently provoked early-stage attacker engagement throughout all simulated lateral-movement cases. Test outcomes indicated that every simulated adversary (100%) contacted at least one deception asset before reaching secured production resources. The mean detection delay when the SMB lure was engaged stayed under one second, allowing swift containment actions prior to extensive spread.

The watchdog-driven self-repair mechanism also showed high robustness against hostile tampering efforts. In simulations where the endpoint agent was terminated, the watchdog successfully reinstated monitoring capabilities, achieving an average restoration time of roughly two seconds. This maintained uninterrupted telemetry collection and reduced gaps in forensic visibility during attempts to bypass security. The test was performed under the premise that the user lacked administrative privileges.

C. LogStamping Scalability and Integrity

The blockchain-enabled forensic audit platform, LogStamping, showed notable scalability gains over traditional blockchain logging designs. Using grouped SHA-256 digest anchoring markedly cut down blockchain storage expansion by lowering the count of on-chain transactions needed for telemetry validation. Benchmarks versus current blockchain forensic solutions such as ProvChain, BCALS, and EngraveChain revealed that the introduced grouped hashing approach trimmed blockchain storage burden by roughly 95% relative to simple per-entry anchoring methods.

With a telemetry load of 10,000 log records, the FORLENS LogStamping system used roughly 0.62 MB of blockchain space, whereas ProvChain required 10 MB and EngraveChain needed 12.5 MB. Even with the smaller blockchain expansion, the forensic integrity check stayed extremely dependable. Tests of integrity rebuilding that included inserting, deleting, changing lines, and reordering events detected tampering with 100 % accuracy in every tested case.

Verification latency stayed within acceptable operational limits despite batch forensic reconstruction procedures. Average forensic verification time for 10,000 telemetry records stayed under 155 seconds, surpassing several existing blockchain-assisted forensic architectures while using considerably less storage.

The lightweight endpoint monitoring architecture showed minimal computational load during continuous operation. Average CPU usage stayed between 2% and 4% while performing active telemetry gathering, behavior analysis, deception monitoring, blockchain synchronization, and orchestration processing. Memory consumption remained within 60 MB to 90 MB, varying with telemetry intensity and active machine-learning inference workloads. These findings verify the suitability of the FORLENS framework for deployment in resource-constrained SME environments using commodity hardware.

D. Operational Usability and Resource Efficiency

The Explainable Artificial Intelligence (XAI) module markedly enhanced operational usability and the efficiency of administrators' responses. A controlled usability study with non-technical administrators showed that readable behavioral narratives greatly boosted operators' comprehension of identified threats. In the absence of XAI support, the average accuracy of incident interpretation hovered around 51%, whereas explanations aided by XAI raised it to roughly 89%.

Moreover, adding easy-to-understand behavioral explanations cut the Mean Time To Respond (MTTR) from around 30 minutes to less than 5 minutes in simulated security incidents. Administrators said their confidence rose markedly when handling behavioral alerts that included contextual explanations and suggested containment steps.

The self-directed orchestration and response system also showed strong containment ability during ransomware and lateral-movement drills. High-confidence alerts automatically initiated process kill, file sequestering, network isolation, SMB session shutdown, IOC blocking, and blockchain forensic sync workflows. The orchestration flow kept response times short while retaining admin supervision via confidence-based escalation rules.

An assessment compared to current enterprise and open-source security frameworks uncovered numerous strengths of the FORLENS ecosystem. Vendor solutions like CrowdStrike Falcon and Microsoft Sentinel offer sophisticated behavioral analytics yet depend heavily on cloud infrastructure, involve licensing expenses, and demand a dedicated SOC team. Community-driven tools such as Wazuh and OSSEC present more budget-friendly options but miss built-in blockchain forensics, deception-enhanced behavior scoring, Explainable AI, and a lightweight autonomous orchestration.

FORLENS showed an uncommon equilibrium among forensic integrity, behavioral intelligence, operational explainability, deception-assisted detection, autonomous response, and computational efficiency. The framework effectively translated advanced enterprise cybersecurity concepts into a lightweight architecture appropriate for SME operational environments.

V. LIMITATIONS AND FUTURE WORK

A. Limitations

Despite the promising results, several limitations were identified during the evaluation. First, blockchain synchronization delays increased slightly during large-scale telemetry spikes with high event-generation rates. Although grouped hashing significantly reduced storage growth, future large-scale deployments may require optimized distributed validators and adaptive synchronization scheduling.

In addition, the FORLENS platform was evaluated mainly in simulated and controlled environments. Its performance may therefore differ under real-world deployment conditions. Most technologies used in the framework are free and open-source solutions, which may not provide the same level of robustness, scalability, or vendor support as commercial alternatives. As a result, practical enterprise deployments may require more advanced or commercial-grade tools to operate effectively in high-noise environments.

B. Future Work

Behavioral analysis methods based on machine learning remain vulnerable to evasion and poisoning attacks. Advanced adversaries capable of gradually modifying behavioral patterns may influence adaptive baseline learning mechanisms over time. Future research should therefore investigate adversarially robust machine learning techniques and federated anomaly detection architectures.

Furthermore, the current experimental evaluation focused primarily on Windows-based SME environments. Additional validation across enterprise-scale infrastructures, Linux platforms, hybrid cloud systems, and Industrial IoT environments would improve the generalizability and applicability of the framework.

VI. CONCLUSION

This research presented FORLENS, a lightweight AI-driven endpoint security ecosystem designed specifically to address the growing cybersecurity challenges faced by Small and Medium Enterprises (SMEs). The study identified that existing enterprise-grade security solutions frequently remain inaccessible to SMEs due to high infrastructure costs, operational complexity, excessive computational requirements, and dependence on dedicated Security Operations Centers (SOCs). At the same time, conventional security mechanisms such as signature-based antivirus systems and centralized logging architectures remain insufficient against modern adversarial techniques including ransomware, fileless malware, credential theft, insider abuse, and lateral movement attacks [72], [74], [75].

To overcome these constraints, the FORLENS architecture merges several cutting-edge cyber-security functions within a single, resource-lean design. It brings together blockchain-supported forensic audits, low-overhead deception engineering, hybrid AI-based behavior analytics, Explainable Artificial Intelligence (XAI), self-directed orchestration workflows, and robust endpoint monitoring to create an all-encompassing defense ecosystem tailored for SME operations.

An important outcome of this study was the creation of the LogStamping forensic framework, which incorporated grouped SHA-256 blockchain anchoring to enable scalable, tamper-proof forensic preservation. By consolidating telemetry into collective digests prior to blockchain synchronization, the framework markedly lowered blockchain storage demands while still offering cryptographically provable integrity. Experimental results showed that the grouped anchoring approach delivered considerable scalability gains over traditional blockchain logging designs, while still supporting dependable forensic reconstruction.

The behavioral analytics engine boosted endpoint visibility by incorporating Isolation Forest anomaly detection, LSTM autoencoder sequential modeling, OSINT threat-intelligence enrichment, and deception-driven contextual scoring. The hybrid analytical framework reliably detected ransomware processes, irregular PowerShell activity, credential misuse, lateral-movement

actions, and dubious behavior patterns, all while keeping computational load low enough for typical SME hardware.

Integrating lightweight SMB deception techniques gave a proactive early-warning function against lateral movement efforts without needing a dedicated honeypot setup. Moreover, the watchdog-driven self-healing framework enhanced durability against adversary tampering by guaranteeing constant surveillance and swift recovery of disabled endpoint agents.

The Explainable Artificial Intelligence component tackled the operational trust shortcomings typically linked to black-box AI models by producing human-readable behavior narratives and contextual response advice. Experimental testing showed that XAI-guided orchestration markedly boosted administrator comprehension, lowered alert fatigue, and cut Mean Time To Respond (MTTR) in simulated incident-response scenarios.

Experimental findings verified that FORLENS effectively harmonized detection precision, forensic soundness, operational clarity, scalability, and resource efficiency. The system kept CPU and memory usage low while delivering robust anomaly detection and dependable tamper-detection abilities. Further comparative tests showed that FORLENS offers a distinctive mix of lightweight operation, behavioral insight, deception-aided detection, blockchain-supported forensic audit, and autonomous orchestration rarely seen in current SME-focused security solutions.

Even with these advances, a number of constraints persist. Deploying at an enterprise scale can add greater synchronization overhead and complicate telemetry management. Moreover, anomaly detection powered by machine learning continues to be susceptible to adversarial manipulation and poisoning attacks. Consequently, upcoming investigations ought to concentrate on federated learning, adversarially robust behavioral analytics, distributed validator optimization, privacy-preserving AI inference, and edge-centric orchestration mechanisms.

Overall, this study shows that sophisticated enterprise cybersecurity ideas can be effectively transformed into a cost-effective, scalable, transparent, and operationally feasible framework appropriate for SME settings. By combining blockchain forensics, AI-based behavioral analytics, deception engineering, and autonomous response within a single architecture, FORLENS helps narrow the cybersecurity capability divide between large corporations and limited-resource organizations while enhancing resilience to contemporary endpoint threats.

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